Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Python engineer with 2+ years building **scalable applications, REST APIs, web services, automation scripts, and data-processing pipelines** using FastAPI, Flask, PostgreSQL, MySQL, AWS S3, Linux, Docker, Git, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Developed and maintained Python FastAPI services processing 5,000+ monthly data jobs, implementing validation, reconciliation, error handling, and well-documented CI/CD releases while sustaining 99%+ reliability in production at scale.
- Designed PostgreSQL and SQL Server data models supporting 10,000+ records, writing efficient SQL transformations and reducing application latency 35% through indexing, query tuning, debugging, and performance analysis.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Built Python and Flask REST APIs on AWS handling 10,000+ daily requests, integrating third-party OpenAI, Anthropic, and Qwen services with secure ingestion, fallback logic, and monitoring at scale.
- Implemented multithreaded data processing with ThreadPoolExecutor for JSON, PDF, DOCX, ZIP, text, and image inputs, processing 1,000+ files per batch with fault-tolerant execution and automated validation at scale.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed clean, modular Python and PyTorch pipelines processing 100,000+ trajectory records, improving long-horizon planning 18% through automated preprocessing, reproducible tests, debugging, and systematic performance evaluation at scale.
- Maintained configuration, experiment tracking, and technical documentation using Hydra and Weights & Biases, collaborating on system design and evaluation reviews while improving closed-loop software stability 15% during simulation testing.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed Python data-processing pipelines using AWS Glue and S3 to transform JSON, CSV, and Parquet into normalized formats, increasing ETL efficiency 68% and reducing manual preparation at scale.
- Maintained Git-based Jenkins CI/CD across Linux, Windows, and macOS, executing 20+ unit and integration tests with PyTest and Mockito while improving API response times 30% at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built Node.js REST services with Azure SQL, Docker packaging, and Azure deployment, supporting two million synchronized records across eight modular application services and business functions at scale.
- Expanded automated coverage with Selenium, Jest, and Mocha across unit, integration, and end-to-end workflows, achieving 95%+ build validation while reliably reducing deployment time 50% through Jenkins CI/CD.

Skills

Python: Python, FastAPI, Flask, Django, automation scripting, data processing, multithreading
APIs and Data: REST APIs, web services, JSON, CSV, Parquet, SQL, PostgreSQL, MySQL
Cloud: AWS, S3, Glue, RDS, Azure, GCP, Docker, serverless architecture, cloud storage
Testing: PyTest, Selenium, unit testing, integration testing, automated test cases, validation
Engineering: Git, GitHub Actions, CI/CD, Jenkins, Linux, Unix, Agile/Scrum, debugging, documentation

Projects

Linguistix | *Python, REST API, AWS, Docker*

- Developed a containerized Python REST API on AWS CodePipeline and ECS Fargate with automated CI/CD releases and validation checks, achieving 87% measured accuracy and reliable production-grade performance.

CaptionCraft - Image Caption Generator Live | *Python, Flask, Testing*

- Built scalable Flask inference APIs and benchmarked BLIP, GIT, and ViT-GPT-2 across 10,000 images and 100,000 captions, reliably improving measured caption accuracy 15% through systematic model testing.

Big-Brain Live | *Data Pipeline, Serverless Application*

- Engineered a serverless cloud data ingestion and indexing pipeline processing 50,000+ records, implementing incremental updates, metadata persistence, secure access controls, and performance tuning that reduced retrieval latency 40%.